



# Introduction to Aksha

Advanced AI Solutions for Manufacturing Excellence

---

# Overview of Aksha

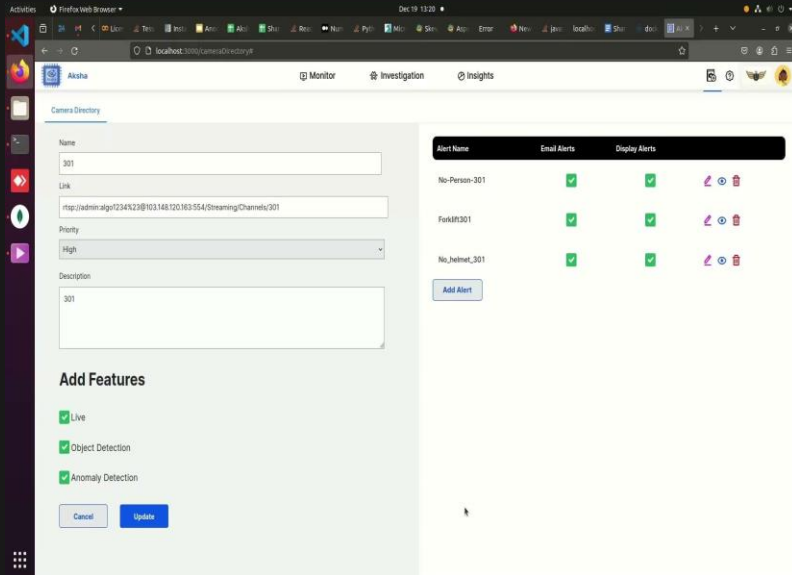


- ▶ Aksha is an intelligent surveillance system designed for automated monitoring with minimal human intervention.
- ▶ Utilizes advanced technologies including Computer Vision, Artificial Intelligence, and Generative AI for enhanced functionality.
- ▶ Capable of detecting objects and unusual situations, enabling proactive responses and analysis.

# Architectural Approaches

- Offers multiple architectural configurations: Aksha on Edge (On Premise), Aksha UI & Backend on Edge, and hybrid models combining Edge & Cloud.
- Supports seamless integration with existing infrastructure to enhance operational efficiency.
- Designed for scalability and flexibility to adapt to various business needs.

# Key Features of Aksha

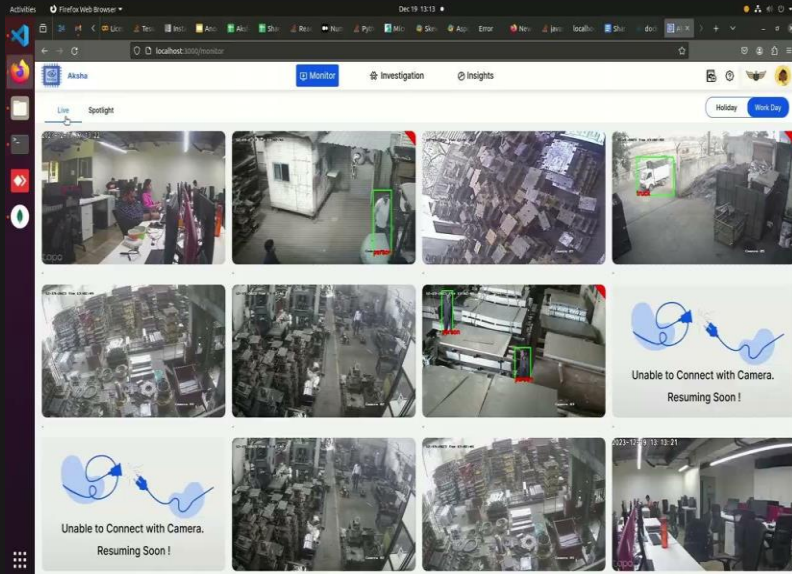


- ▶ Real-time live recording and processing of video data for immediate insights.
- ▶ Advanced object detection capabilities including identification of personnel, vehicles, and safety gear compliance.
- ▶ Customizable alerts allow users to set specific notifications for critical events, enhancing responsiveness.

# Collaboration and Customization

- ▶ In collaboration with NDIL, Aksha has been customized for worker safety, integrating Edge + Cloud solutions.
- ▶ Phased development approach: Phase I focused on foundational capabilities, Phase II enhanced features with Generative AI, and Phase III optimized performance.
- ▶ Successful implementation on Raspberry Pi 4B demonstrates versatility and cost-effectiveness.

# Customer Testimonials



- ▶ Clients report significant reductions in human monitoring efforts and enhanced detection of abnormal events.
- ▶ Feedback highlights the value of Aksha in transforming traditional CCTV setups into proactive safety systems.
- ▶ Continued interest in future updates indicates strong customer satisfaction and trust in Aksha's capabilities.

# Scalability and Future Potential

- ▶ Effortless scalability through microservices architecture using Kubernetes and Kafka for high message volumes.
- ▶ Decoupled architecture enhances fault tolerance and maintainability, allowing for independent service updates.
  - ▶ Real-time data processing capabilities ensure timely insights, positioning Aksha as a leader in intelligent surveillance solutions.